Housing Growth, Neighborhood Inequality, and Zoning Capacity in San Francisco

Part 2: Midterm Report

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Github Link: https://github.com/NihilAkatsuki/sf-housing-growth

1. Research Question

Problem Statement

How did San Francisco's neighborhood housing growth vary by income and renter share from 2005–2025, and how does the Family Zoning Plan (effective January 12, 2026) distribute added capacity by neighborhood income and past housing growth?

Subaspects of Problem Statement

1. Which San Francisco neighborhoods experienced the highest and lowest housing growth rates relative to their initial housing stock from 2005–2025?
2. How did housing growth rates from 2005–2025 vary by neighborhood income and renter share?
3. How is additional housing capacity under the Family Zoning Plan distributed across neighborhoods with different levels of income and historical housing growth?

Since our initial proposal, we replaced percentage housing growth rates with absolute net additions because the available Land Use dataset does not provide a reliable 2005 housing-stock baseline. We also added San Francisco Department of Building Inspection (DBI) data to locate housing records unmatched through Land Use parcel identifiers (initially about 40% of records). We matched these records by building permit identifier (BPA), using primary-address coordinates only when a permit had one unique location and preserving existing Land Use matches. This recovered 2,005 records, reducing unlocated records from 2,027 to 22 and improving coverage of neighborhood housing growth. Our current analysis focuses on historical housing production, while the Family Zoning Plan analysis remains a planned next phase.

2. Data Cleaning Process

Housing Production

In the Housing Production dataset, we converted the first completion date to datetime format and kept records from 2005–2025 to align with our research question. We removed records with missing net changes because their contribution to housing growth is unknown, and zero net changes because they represent no additions or losses. We retained negative changes to account for housing removals when calculating net growth. Finally, we renamed six columns for readability and consistency (e.g., "bpa" to "BPA" and "blocklot" to "BlockLot") and saved the cleaned dataset as `housing_production_filtered.csv`, containing 5,981 records and 57,982 total net units completed.

San Francisco Land Use Data

We preserved "mapblklot" as text to retain leading zeros and converted residential_units into numeric values. We also reviewed missing parcel identifiers and resolved duplicate parcel records

using previously audited residential-unit values. These steps reduced the dataset from 153,765 to 153,754 records, removing 11 conflicting or duplicate entries. For geographic matching, we retained parcel identifiers, centroid coordinates, residential-unit counts, and geography types. We used Land Use centroids to locate housing permits rather than treating current residential-unit counts as a 2005 housing-stock baseline as originally planned.

DBI Locations

Of 7421 raw DBI records, we retain 5955 (80.2%) with primary_address_flag = "Y" and usable longitude-latitude coordinates. The remaining 1466 records (19.8%) are excluded from the DBI lookup, not from Housing Production. This ensures that only primary addresses with available coordinates are used for geographic matching. We remove identical permit-coordinate records and reject permits with multiple distinct locations to prevent ambiguous matches and duplicate housing counts. Neither check removes additional candidates in this dataset. Using BPA to match DBI with Housing Production, we recover 2005 previously unmatched records, leaving only 22 records (826 net housing units) without coordinates.

Median Household Income — B19013 (ACS 2020–2024)

We skip the descriptive second CSV row and filter GEO_ID values beginning with "1400000US06075" to retain only San Francisco Census Tracts. We extract the last 11 characters as GEOID and remove duplicate tract identifiers to prevent repeated matches. We clean B19013_001E by removing commas and plus signs before converting values to numeric format. Invalid estimates become missing, while "250000+" is retained as a lower bound for classification rather than treated as an exact income. Of 244 Census Tracts, 234 have valid income estimates. The other 10 remain unclassified, preserving their 1571 net housing additions rather than excluding them from the overall housing analysis.

Tenure — B25003 (ACS 2020–2024)

We skip the descriptive second CSV row, filter San Francisco Census Tracts using GEO_ID, extract the 11-digit GEOID, and remove duplicate tract identifiers. We use B25003_001E for total occupied housing units and B25003_003E for renter-occupied units(Renter Share = Renter-Occupied Units / Total Occupied Units.) We exclude tracts with zero occupied units to avoid division by zero and require renter counts to fall between zero and the total occupied units. Four tracts are excluded from the renter-share calculation, leaving 240 valid tracts. After merging B25003 with B19013, 234 tracts have both valid measures and are classified into three equal-sized income and renter-share groups using pd.qcut(). Temporal limitation: ACS classifications reflect 2020–2024 neighborhood conditions, while housing production covers 2005–2025. We therefore use these classifications for descriptive comparisons, without assuming neighborhood characteristics remained constant over time.

3. Methods & Techniques

Step 1: Prepare inputs. We cleaned all of our datasets as specified in the section above.

Step 2: Assign geographic locations to housing records. We assign latitude and longitude to each Housing Production record using three matching approaches. First, we left join Housing Production's "BlockLot" with Land Use's "mapblklot" for individual parcels, locating 3,938 records. Second, we extract individual parcel IDs from multi-parcel Land Use records and match them to housing records, locating 16 additional records using shared group centroids. We exclude incomplete parcel lists and ambiguous matches. Finally, for remaining unmatched records, we match Housing Production's "BPA" with DBI's "permit_number", using only primary-address records with valid coordinates and exactly one unique location. This recovers 2,005 additional records without replacing existing Land Use coordinates. Overall, we locate 5,959 of 5,981 records, leaving 22 unmatched records representing 826 signed net units.

Step 3: Find the tract containing each point. We convert housing coordinates into geographic points and use GeoPandas spatial joins (sjoin, with the within predicate) to assign each point to a 2020 Census Tract using its [GEOID]. Both points and tract polygons use WGS84 (EPSG:4326). All 5,959 geolocated records are assigned to tracts without duplication. The remaining 22 records are preserved with missing [GEOID] values.

Step 4: Sum housing changes by tract. We group records by "GEOID" and sum Net Units Completed, retaining both positive additions and negative changes. We also calculate separate totals for housing additions and losses, along with the number of production records per tract. The resulting "net_units_added_2005_2025" represents net housing additions for each tract. All 244 Census Tracts are retained, with zero assigned where no housing production was successfully located.

Step 5: Calculate and map net additions. We visualize "net_units_added_2005_2025" using a census tract choropleth map. Across the full sample, total net housing additions equal 57,982 units, of which 57,156 are assigned to tracts. The remaining 826 units have no tract assignment. These results represent absolute net additions rather than percentage growth rates, which require a historical housing-stock baseline.

Step 6: Classify recent neighborhood characteristics. Using ACS income and tenure data, we calculate renter share as renter-occupied units divided by total occupied units. We merge the two datasets by [GEOID] and use qcut to classify tracts into three equal-sized groups (low, medium, and high) based on median household income and renter share. Among 234 tracts with valid data for both measures, each classification contains 78 tracts per group. Tracts with missing socioeconomic information remain unclassified.

Step 7: Compare group totals and shares. We separately group tracts by income level and renter share, calculating the number of tracts, housing production records, and total net housing additions in each group. We then calculate each group's share of citywide net additions as:

$$Share\ of\ Citywide\ Net\ Additions\ = \frac{Group\ Net\ Units}{57{,}982} \times 100\%$$

Unclassified tracts and housing records without geographic assignments are accounted for separately. These comparisons allow us to examine how housing production varies across neighborhoods with different income levels and renter shares.

4. Early Results

Our early results show that housing production was highly concentrated in particular parts of San Francisco.

The choropleth map (Figure 1) identifies substantial concentrations in eastern San Francisco, while many western and central tracts recorded relatively small net additions.

The income comparison (Figure 2, left panel) shows that high-income tracts recorded 22,495 net additions (38.8% of citywide net additions), compared with 17,528 (30.2%) in middle-income tracts and 15,562 (26.8%) in low-income tracts. The renter-share comparison (Figure 2, right panel) reveals a larger difference. High-renter-share tracts accounted for 27,660 net additions (47.7%), compared with 20,256 (34.9%) in middle-renter-share tracts and 7,669 (13.2%) in low-renter-share tracts.

The histogram (Figure 3) shows a strongly right-skewed distribution: most tracts experienced modest increases, while a small number contributed substantially more housing. The ranking of the ten tracts with the largest net additions (Figure 4) identifies Census Tract 06075060701 as the highest, with 4,251 net units.

One surprising result we found was that high-renter-share tracts had only 1,416 Housing Production records but 27,660 net additions, whereas low-renter-share tracts had 2,431 records but only 7,669 net additions. This indicates that the number of housing units represented by each production record varies a lot across groups, suggesting that larger residential developments may have played a greater role in high-renter-share neighborhoods.

We expected housing construction to be geographically uneven, but the magnitude of the differences between renter-share groups was notable. It is also important to note that the ACS groups describe neighborhoods during 2020–2024, whereas housing production spans 2005–2025. This means that we cannot assume that income or renter share remained stable during those two decades or infer that these socioeconomic characteristics caused the observed production patterns. Furthermore, we measured absolute net additions rather than growth rates, so differences may partly reflect neighborhood size and existing development patterns.

5. Plan to Finish

Remaining Work

Our next priority is to complete Step 8, analyzing how the Family Zoning Plan changes housing development opportunities across San Francisco.

We have already collected three zoning datasets from the city's ArcGIS application: height controls (92,086 records), zoning-use districts (10,302 records), and base-density regulations (154,464 records). Next, we will use parcel identifiers (`mapblklot`) and spatial joins to connect these

datasets to census tracts. We will compare previous and adopted zoning controls to identify changes in building height, permitted land use, and density regulations. We will then aggregate these changes by tract and compare their distribution across neighborhoods with different income levels, renter shares, and historical housing growth.

Finally, we plan to create an interactive Plotly map to visualize the geographic distribution of zoning changes. If sufficient information becomes available, we will also estimate additional housing capacity under the new regulations.

Timeline and Responsibilities

By November 8 (Colton): Finalize tract-level historical housing growth data and integrate ACS income and renter-share variables.

By November 15 (Tanushri): Complete historical housing growth comparisons across income and renter-share groups, including summary statistics and key findings.

By November 22 (Luoqi): Merge the Family Zoning Plan height, zoning-use, and density datasets with parcel and census tract data, identifying changes in zoning regulations.

By November 30 (Colton): Complete the Family Zoning Plan analysis, comparing zoning changes across neighborhoods with different income levels, renter shares, and historical housing growth.

By December 7 (Tanushri): Develop an interactive Plotly map and supporting visualizations showing the geographic distribution of zoning changes and their relationship to neighborhood characteristics.

By December 12 (Luoqi): Integrate the findings and visualizations, validate the results, and start to complete the final report draft.

By December 15 (All): Review, revise, and submit the final project report and supporting code.

Potential Challenges

Limited Data Availability: The zoning datasets do not directly provide the number of additional housing units allowed under the plan, so additional information or assumptions may be required to estimate housing capacity.

Causal Attribution: We cannot determine whether housing built after January 2026 was directly caused by the Family Zoning Plan, so we will focus on changes in zoning regulations rather than attributing completed construction to the policy.

Incomplete Zoning Coverage: Some zoning layers do not cover every parcel in San Francisco, which may limit comparisons across neighborhoods. We will document unmatched parcels and avoid treating missing data as unchanged zoning.